30 December 1975
SUPERSEDING
MIL-S-0043243B(GL)
3 September 1974 and
MIL-S-43243A
10 January 1967

MILITARY SPECIFICATION

SEPARATOR, PARACHUTE REMOVABLE CONNECTOR LINK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size parachute removable connector link separator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup
PPP-T-60	-	Tape: Packaging, Waterproof
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 1670

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MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets

DRAWINGS

ARMY NATICK DEVELOPMENT CENTER

11-1-176 - Separator, Parachute Removable Connector Link; Assembly and Details
11-1-240 - Spring, Extension

(Miniature copies of Drawings 11-1-176 and 11-1-240, identified as figures 1 and 2 respectively, are attached for information only.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the separator to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.2 Materials and construction. The materials, construction, and supplementary processes involved in the fabrication of the separator assembly shall conform to the requirements specified or shown on Drawings 11-1-176 and 11-1-240.

3.3 Workmanship. The Parachute Removable Connector Link Separator, including all components and assemblies shall be fabricated and finished in a thoroughly workman like manner. Particular attention shall be given to the following:

a. All areas shall be free of rust or corrosion, dirt, oil, grease or other foreign matter and tool marks or breaks through plating to metal.

b. Free from metal slivers.

c. All welding, silver soldering, and brazing shall be smooth, clean, free of flux, pits, slag or spatter, porosity and shall be continuous, adherent, and not burnt through. The joints shall be thoroughly fused.

d. Rivets shall be fully peened with heads in full contact with riveted members. The rivets shall be firmly set and completely fill the hole. Rivet heads shall be full, neatly made and concentric with rivet holes. Holes for rivets shall be accurately punched or drilled and free of any obstruction.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required, it shall be examined for the defects in table I and for compliance with dimensions on drawings specified in 3.2. The presence of any defect listed in table I or any dimension not within specified requirements shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

a. Heat treatment of spring (see Drawing 11-1-240).
b. Cementing of felt pad (see Drawing 11-1-1-176).

4.4.3 End item inspection. The inspection lot shall be all separators offered for inspection at one time. The sample unit shall be one complete separator assembly.

4.4.3.1 Visual examination. Examination shall be made of the separator for the defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish, protective	Any area of rust or corrosion	X	
	Any area of no finish, tool marks or breaks through plating to metal		X
	Dirt, oil, grease or other foreign matter		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Design construction and workmanship	Any sharp edges, burrs or metal slivers	X	
	Any component missing or damaged	X	
	Any steel component not assembled or joined as specified	X	
	Felt pad not securely cemented to base		X
	Welding incomplete or improper type	X	
	Weld, braze, or silver solder not smooth, any slag or spatter or burn through		X
	Weld, braze, solder not free of pits or porosity		X
	Weld, braze, solder area not clean or free of flux		X
	Lever binds, not self-returning	X	
	Rivet not firmly set or adequately peened	X	
	Rivet heads not in full contact with adjacent member		X
	Rivet heads not concentric to rivet holes		X
Identification marking	Missing, incomplete; illegible, incorrect, or not applied as specified		X

4.4.3.2 Dimensional examination. Inspection shall be made of the separators to determine compliance with dimensions specified. Any dimension not within specified tolerances shall be classified a defect. The inspection shall be level II with an AQL of 6.5, expressed in terms of defects per hundred units.

4.4.3.3 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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TABLE II. Examination of preparation for delivery

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling Bulged or distorted containers
Content	Number of packages per shipping container is more or less than required

4.4.3.4 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with table III. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 with an AQL of 6.5, expressed in terms of defects per hundred units.

TABLE III. Examination for palletization

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each separator shall be packaged in a folding paperboard box conforming to variety 2, style III, type G, class 1 of PPP-B-566; or setup paperboard box conforming to type I, variety 2, class A, style 4 of PPP-B-676. Inside dimensions of each paperboard box shall be 7-1/8 inches in length, 3-5/8 inches in width and 2-1/8 inches in depth. Each box shall be securely closed with 2 inch minimum width pressure-sensitive tape conforming to PPP-T-60 or PPP-T-76. The tape shall be applied across the center of the length opening and along the bottom and up each side at least one-half the depth of the box.

5.1.2 Level C (commercial packaging). Each separator shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Fifty separators, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636 or overseas type, style A or I, grade A or B, type 2 load of PPP-B-601. Level A packages shall be packed flat, five in length, two in width and five in depth within a shipping container. Inside dimensions of each shipping container shall approximate 18-3/4 inches in length, 14-1/2 inches in width and 11-1/4 inches in depth. Approximate dimensions are furnished as a guide only. Each fiberboard container shall be closed in accordance with method III and reinforced as specified in the appendix of PPP-B-636. Each wood container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B. Fifty separators, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A packages shall be packed flat, five in length, two in width and five in depth within a shipping container. Inside dimensions of each shipping container shall approximate 18-3/4 inches in length, 14-1/2 inches in width and 11-1/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

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5.2.3 Level C (commercial packing). Separators, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization. When specified (see 6.2), separators, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The separator assembly covered by this specification such as MS22002 and MS20118 used to disengage the leg members of mating parachute removable connector links to facilitate the assembly or disassembly of the applicable parachute components used with the connector links.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3 and 6.3).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting

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officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Custodians:

Army - GL
Navy - AS

Review activity:

Army - AV

User activity:

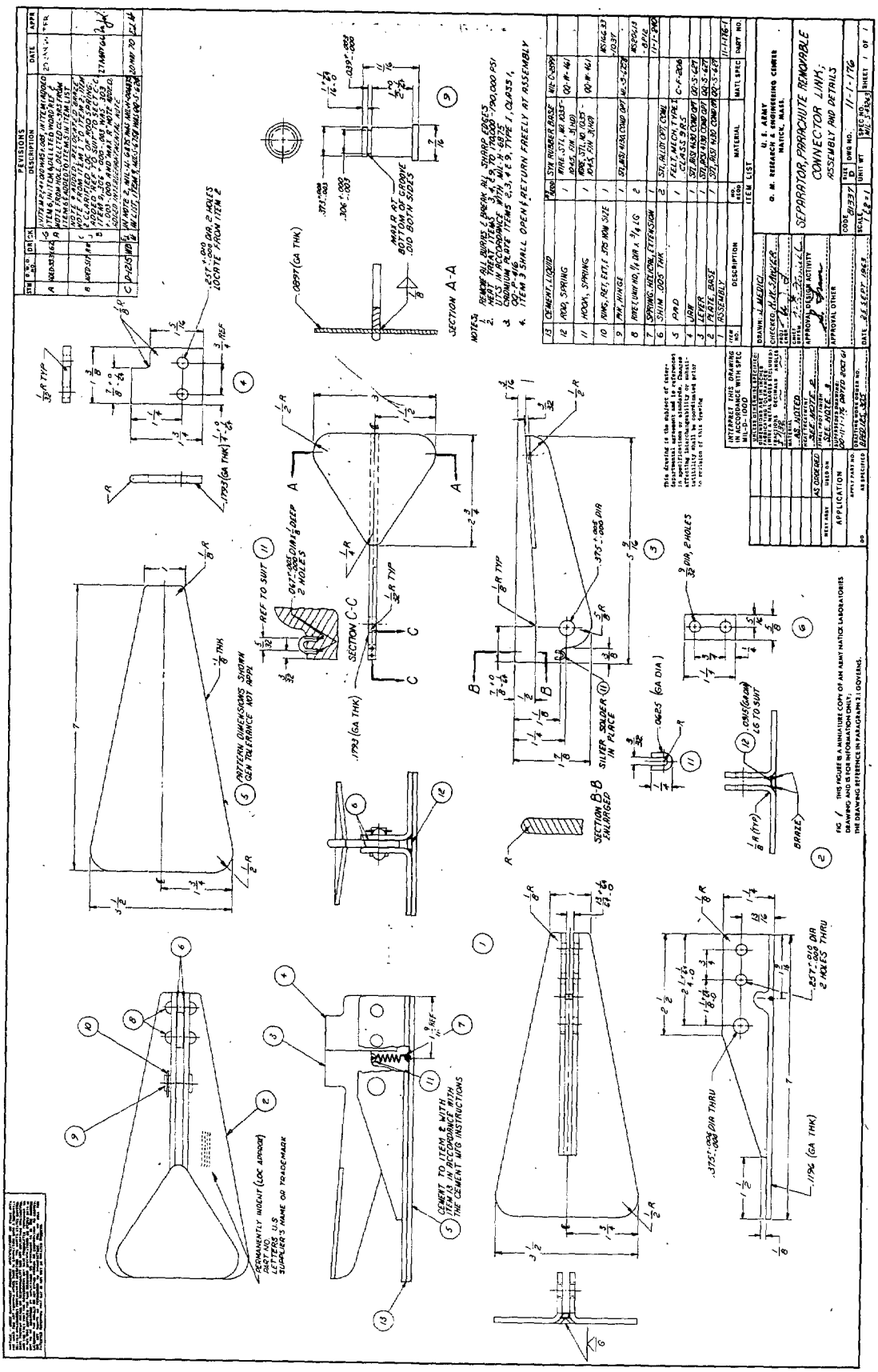
Navy - MC

Preparing activity:

Army - GL

Project No. 1670-0563

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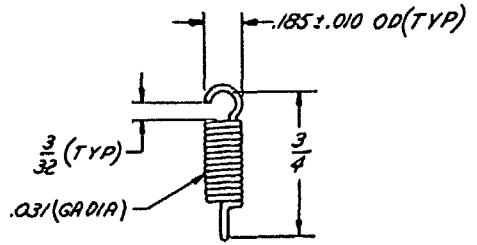


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REVISIONS						
SYM	D.W.O. NO.	DR	CK	DESCRIPTION	DATE	APPR
A	ADED 333	GZ	JA	TYI, CL1 WAS TYII, CL2	20 JAN 64	TFR
B	ADED 517	RW	JB	ADDED (TYP) TO DIM ON SPRING; CHANGED NOTE 5, ADDED INTER-DEPARTMENTAL NOTE	27 MAY 64	
C	ADEL 698	VAQ	FE	3/4 WAS 5/8	20 MAR 68	AJD

NOTES:
1. DIRECTION OF HELIX OPTIONAL
2. TOTAL COILS 13 1/2 ± 50°
3. RELATIVE POSITION OF ENDS 90° ± 20°
4. INITIAL TENSION OPTIONAL
5. MACHINE HOOK BOTH ENDS AS SHOWN



11

Use of drawing is the subject of interdepartmental agreement and is reference in specifications or standards. Changes affecting interchangeability or substitutability shall be coordinated prior to revision of this drawing.

FIG 2 THIS FIGURE IS A MINIATURE COPY OF AN ARMY NATICK LABORATORIES DRAWING AND IS FOR INFORMATION ONLY; THE DRAWING REFERENCE IN PARAGRAPH 2.1 GOVERNS.

11-1-176	
NEXT ASSY	USED ON
APPLICATION	
DO NOT	APPLY PART NO.
~~CO~~	~~AS SPECIFIED~~

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
FABRICATING TOLERANCES:
(WIRE & METAL GAGES EXCLUDED)
FRACTIONS 1/32 DECIMALS ~ ANGLES ~

MATERIAL: QQ-W-470
HEAT TREATMENT: NORMALIZE AT 550F - 15 MIN
FINAL PROT FINISH: TYI, CL1, QQ-P-416
SUPERSEDES DRAWING:
DRAFTING WORK ORDER NO.: ADED 305

DRAWN: J. MERICI
CHECKED: H. W. SAWYER
PROJ ENGR: [illegible]
CHIEF DRAFTSMAN: [illegible]
APPROVAL DESIGN ACTIVITY: S. Train
APPROVAL OTHER:
DATE: 25 SEPT 1969

U. S. ARMY
Q. M. RESEARCH & ENGINEERING CENTER
NATICK, MASS.

SPRING, HELICAL, EXTENSION

CODE 81337 | SIZE B | DWG NO. 11-1-240
SCALE 2:1 | UNIT WT | SPEC NO. MIL-S-43243 | SHEET 1 OF 1

MIL-S-43243C

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-43243C- SEPARATOR, PARACHUTE REMOVABLE CONNECTOR LINK

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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